

# Multiplying and Dividing in Scientific Notation

## Lesson Plan

### 1. Lesson Information

|        |                                                                     |              |                                            |
|--------|---------------------------------------------------------------------|--------------|--------------------------------------------|
| Course | Grade 7 Accelerated<br>(1205050)                                    | Unit         | 4 — Solving Problems with Rational Numbers |
| Lesson | Lessons 2–3 —<br>Multiplying and Dividing<br>in Scientific Notation | Date         | _____                                      |
| Length | 53 minutes (one period)                                             | Benchmark(s) | MA.8.NSO.1.5                               |

### 2. Standards

**MA.8.NSO.1.5** Add, subtract, multiply and divide numbers expressed in scientific notation with procedural fluency.

**Must.** Multiply and divide numbers written in scientific notation with procedural fluency, and write every answer with a coefficient at least 1 and less than 10.

**Must not.** No within-2 limit applies here — that clarification is about addition and subtraction only, and the contrast with 4.01 is the point of the day. Significant digits are 4.04.

### 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR            | What it asks                                                                        | The evidence in this lesson                                                                                                                                   |
|----------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MA.K12.MTR.5.1 | Use patterns and structure to help understand and connect mathematical concepts.    | Notes I — the product is regrouped before it is evaluated, so 'multiply the fronts, add the backs' is read off the structure rather than memorised as a rule. |
| MA.K12.MTR.3.1 | Complete tasks with mathematical fluency.                                           | Boards 1, 3 and 7 — one move each, written and up on the cue; dividing in scientific notation was the lowest single figure on the last state report.          |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | Board 6 — when the room splits between $2 \times 10^{10}$ and $20 \times 10^9$ , which are the same number, the sixty-second re-vote before the reveal.       |

### 4. Learning Target and Essential Question

**Learning target.** I can multiply and divide two numbers written in scientific notation, and write the answer in scientific notation.

**Essential question.** Why must the powers of ten match to add, but not to multiply?

**Building on.** 4.01 — the coefficient rule and the finishing step; MA.7.NSO.1.1 and MA.8.NSO.1.3 (3.01–3.03) — the product and quotient of powers laws.

**Working toward.** 4.04 — real-world problems and significant digits; MA.912.NSO.1 — operations with rational exponents in Algebra 1.

## 5. Sequence

*read from the deck; never typed*

| Segment                             | Min       | What the teacher does                                                                        | What students do                           |
|-------------------------------------|-----------|----------------------------------------------------------------------------------------------|--------------------------------------------|
| Settle in; post the learning target | 1         | Post the target; say the 'today' line and nothing else yet.                                  | Copy the target.                           |
| Warm-Up                             | 5         | Two minutes silent, then reveal. One sentence of reteach per question at most.               | Four retrieval questions, alone, no notes. |
| Notes                               | 11        | Teach from the front. Expand once per law; say the OFF line out loud.                        | Copy the notes; answer the checks aloud.   |
| Example 1                           | 3         | Work the example; name the wrong boards before they appear.                                  | Watch, then work the Your Turn alone.      |
| Your Turn                           | 3         | Circulate; reveal and name what a wrong board most likely did.                               | Work it alone; boards up on the cue.       |
| Example 2                           | 3         | Work the example; name the wrong boards before they appear.                                  | Watch, then work the Your Turn alone.      |
| Your Turn                           | 3         | Circulate; reveal and name what a wrong board most likely did.                               | Work it alone; boards up on the cue.       |
| Whiteboards                         | 13        | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three.    |
| Independent Set                     | 6         | Silent. Circulate and mark what you see; do not teach.                                       | Six questions, written, alone.             |
| IXL                                 | 5         | Assign the day's IXL skills.                                                                 | Start the skills; finish tonight.          |
| <b>Total</b>                        | <b>53</b> |                                                                                              |                                            |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (13 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question                                                                                                                                                       | DOK   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Why do the powers of ten have to match before you add, but not before you multiply?                                                                            | DOK 3 |
| Marcus's power of ten is right and Sara's coefficient is right. Which of the two is closer to the correct answer, and does 'closer' mean anything useful here? | DOK 3 |
| A quotient of two positive numbers comes out as $5.3 \times 10^{-3}$ . What does the negative exponent tell you about how the two numbers compared?            | DOK 2 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks                                                                                                                                                                | Answer                 | If the board is wrong                                                                                                                                                                                                                                                                                                                                    |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $(3 \times 10^4)(2 \times 10^5)$                                                                                                                                                   | $6 \times 10^9$        | $6 \times 10^{20}$ — the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception: $(2 \times 10^4)(3 \times 10^5) = 6 \times 10^{20}$ ]; $5 \times 10^9$ — the coefficients were added [8.NSO.1.5: the operation is multiplication]                                                                                                               |
| 2 | A printer makes $4 \times 10^3$ copies per hour. It runs for $2.5 \times 10^2$ hours. Paper costs 3 cents a sheet. How many copies does it make?                                   | $1 \times 10^6$        | anything built from 3 cents — the unneeded figure was used [the question asks for a count, not a cost]; $10 \times 10^5$ — right value, coefficient not less than 10 [B1G-M 8.NSO.1.5 misconception]                                                                                                                                                     |
| 3 | $\frac{9.6 \times 10^8}{1.2 \times 10^3}$                                                                                                                                          | $8 \times 10^5$        | $8 \times 10^{11}$ — the exponents were added [8.NSO.1.5: division subtracts the exponents]; $8.4 \times 10^5$ — the coefficients were subtracted instead of divided [8.NSO.1.5]                                                                                                                                                                         |
| 4 | $\frac{2.65 \times 10^{11}}{5 \times 10^{13}}$                                                                                                                                     | $5.3 \times 10^{-3}$   | $0.53 \times 10^{-2}$ — right value, but 0.53 is not at least 1 [8.NSO.1.5: the coefficient satisfies $1 \leq a < 10$ ]; $5.3 \times 10^{-1}$ — the point moved but the exponent did not                                                                                                                                                                 |
| 5 | $(2 \times 10^4)(3 \times 10^5)$ . Which is the product?                                                                                                                           | $A - 6 \times 10^9$    | (B) the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception: $(2 \times 10^4)(3 \times 10^5) = 6 \times 10^{20}$ ]; (C) the coefficients were added instead of multiplied [8.NSO.1.5: the operation is multiplication]; (D) the exponents were subtracted, $4 - 5$ [8.NSO.1.5: multiplication adds the exponents]                             |
| 6 | $(5 \times 10^6)(4 \times 10^3)$ . Which is the product, written in scientific notation?                                                                                           | $A - 2 \times 10^{10}$ | (B) equal in value, but 20 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving $12 \times 10^9$ instead of $1.2 \times 10^{10}$ ]; (C) the coefficient was repaired but the exponent was not raised [8.NSO.1.5: point one place left, exponent up one]; (D) the exponents were multiplied and the coefficient was left over 10 [8.NSO.1.5] |
| 7 | $\frac{3 \times 10^8}{3 \times 10^{-4}}$                                                                                                                                           | $1 \times 10^{12}$     | $1 \times 10^4$ — the negative sign on the bottom exponent was dropped [8.NSO.1.5: subtracting a negative adds]; $1 \times 10^{-12}$ — bottom minus top [8.NSO.1.5: top minus bottom]                                                                                                                                                                    |
| 8 | $(8.4 \times 10^7) \div (n \times 10^2) = 4.2 \times 10^5$ . What is the value of n?                                                                                               | $n = 2$                | 4.2 — the coefficients were subtracted, $8.4 - 4.2$ [8.NSO.1.5: the coefficients divide]; 0.5 — $4.2 \div 8.4$ instead of $8.4 \div 4.2$                                                                                                                                                                                                                 |
| 9 | For $(7.2 \times 10^9) \div (2.4 \times 10^4)$ , Marcus writes $4.8 \times 10^5$ and Sara writes $3 \times 10^{13}$ . Name what each one did wrong, and give the correct quotient. | $3 \times 10^5$        | 'they both got it wrong' — neither error is named [the item asks what each one did]; $4.8 \times 10^5$ — Marcus's error repeated                                                                                                                                                                                                                         |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

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**ESE / IEP.** Give a two-column organiser — COEFFICIENTS on the left, POWERS OF TEN on the right — and have the student split every expression into the two columns before computing anything. The regrouping step in Notes I becomes a physical act, and the two laws are then applied one column at a time.

**ELL.** Product and quotient are the two nouns, multiply and divide the two verbs. Write them in pairs on the board — PRODUCT goes with MULTIPLY and ADD the exponents, QUOTIENT goes with DIVIDE and SUBTRACT the exponents — and read each board's operation aloud before the boards go up.

**Enrichment.** Ask for  $(4 \times 10^5)^2$  without a calculator ( $1.6 \times 10^{11}$ ), then for the two numbers in scientific notation whose product is  $1.2 \times 10^7$  and whose quotient is  $1.2 \times 10^1$  ( $1.2 \times 10^4$  and  $1 \times 10^3$ ).

## 10. Closure, Vocabulary, Materials, Practice

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**Closure.** Board 9 is written work and it is the exit evidence: the student has to name Marcus's error and Sara's error separately and give  $3 \times 10^5$ . Read the boards, not the papers.

**Vocabulary.** **product of powers law; quotient of powers law; regrouping a product**

**Materials.** Whiteboards and markers. Calculators are allowed on everything; a calculator is the fastest way to settle whether  $6 \times 10^9$  and  $6 \times 10^{20}$  could possibly both be the answer to whiteboard 5.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Multiply numbers written in scientific notation — YZU; Divide numbers written in scientific notation — SGT.